A Near Real-Time ISAC and Localization Platform using srsRAN

Samanvitha Balijepalli
Dept. of ECE
Rice University
Houston, USA
sb273@rice.edu

Aniket Jadhav
Dept. of ECE
Rice University
Houston, USA
aj156@rice.edu

Sandra Mansour
Dept. of ECE
Rice University
Houston, USA
sm367@rice.edu

***Abstract*—Detecting malicious or unauthorized signals such as jammers often requires specialized sensing hardware that is costly to deploy, maintain, and adapt. This work presents a software-based 5G framework for jammer detection and future localization using the srsRAN stack without major hardware modification. The proposed system creates standards-compliant sensing opportunities through data scheduling by reserving quiet symbol intervals within the 5G frame structure, allowing the gNB to observe interference-related measurements during normal communication. In the current stage, jammer activity is detected using communication-quality indicators derived from the scheduled sensing windows, particularly SNR and jammer power, together with lightweight threshold-based classifiers. Experimental results show reliable separation between jammer OFF and jammer ON conditions, with jammer power achieving perfect classification in the evaluated dataset. A MUSIC-based angle-of-arrival baseline is also validated to support the future localization stage. These results demonstrate the feasibility of low-overhead jammer sensing and provide a foundation for future near-real-time localization.**

***Index Terms*—Integrated Sensing and Communication (ISAC), jammer detection, jammer localization, srsRAN, 5G NR, wireless sensing, threshold-based detection, MUSIC**

I. Introduction

Integrated Sensing and Communication (ISAC) has emerged as an important direction for next-generation wireless systems, in which communication infrastructure is reused to support sensing, detection, and localization tasks. By leveraging existing radio resources for both data transmission and environmental awareness, ISAC can improve spectrum efficiency while reducing reliance on separate sensing hardware. This is especially relevant for jammer detection, where conventional approaches often depend on dedicated monitoring equipment that is costly to deploy, maintain, and adapt.

Open and programmable wireless platforms provide an attractive foundation for developing flexible jammer-sensing systems. In parallel, high-resolution parameter estimation methods such as MUSIC [1] and structured-measurement approaches such as SpotFi [2] have shown that communication-derived signals can support localization and environmental inference without requiring specialized sensing devices. These advances motivate the use of open-source 5G software stacks as experimental platforms for integrated sensing and communication.

This work presents a software-based framework for jammer detection and future localization using the srsRAN [3] stack within an open 5G communication environment. Rather than relying on dedicated sensing signals or external RF instrumentation, the proposed system creates standards-compliant sensing opportunities directly from the communication process by reserving quiet symbol intervals within the 5G frame. As illustrated in Fig. 1, the current phase focuses on jammer detection using communication-quality indicators such as SNR and jammer power, while the future phase extends the framework toward localization using richer structured signal features and the MUSIC algorithm.

The main contribution of this work is the design and evaluation of a low-overhead scheduler-based sensing framework for jammer detection in an open-source 5G platform. Experimental results show that the scheduled sensing measurements clearly distinguish jammer OFF and jammer ON conditions, and that jammer power provides a highly discriminative first-stage detection feature. These results establish a practical foundation for extending the framework toward near-real-time jammer localization in future work.

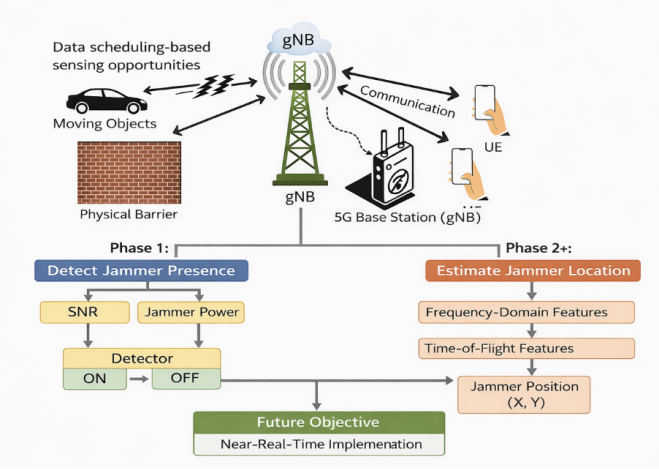


Fig. 1. Joint communication and sensing framework for jammer detection and localization using data-scheduling-based sensing opportunities in a 5G gNB.

II. Methodology

A. System Architecture

The proposed system is built on the srsRAN 5G software stack and uses communication-driven sensing opportunities to detect and eventually localize jammer activity without requiring dedicated sensing hardware. Instead of relying on standalone sensing signals, the framework modifies the gNB scheduler to create periodic quiet windows within the 5G NR frame. During these intervals, the gNB transmitter is silent and the receiver captures ambient RF energy, interference, and jammer-induced signal components. This design remains compliant with 3GPP TS 38.211 and TS 38.213 specifications, ensuring compatibility with standard 5G NR operation.

The full platform consists of an Open5GS core network, a modified srsRAN gNB, an srsUE instance, and a jammer proxy inserted into the ZeroMQ signal path. The gNB and UE maintain normal communication during standard transmission periods, while the jammer proxy injects controlled continuous-wave interference after a configurable delay. To support sensing, the scheduler blanks symbols 12 and 13 in every tenth slot, creating a low-overhead quiet window in which the receiver directly observes the channel. IQ samples from these windows are stored for post-processing, while the jammer proxy simultaneously logs UE power, jammer power, and signal-to-interference ratio values.

A post-processing layer converts the captured IQ data and signal logs into MATLAB-compatible data structures for jammer detection. In the current stage, sensing is performed using communication-quality indicators derived from the scheduled sensing windows, particularly SNR and jammer power. These measurements are processed using lightweight threshold-based detectors to determine whether jammer activity is present. The longer-term goal is to extend this first-stage detection pipeline toward localization using richer signal features and the MUSIC algorithm [1], consistent with prior structured-measurement localization approaches such as SpotFi [2]. As a result, the proposed architecture provides a unified experimental platform for low-overhead jammer sensing, detection, and future localization within an open-source 5G framework.

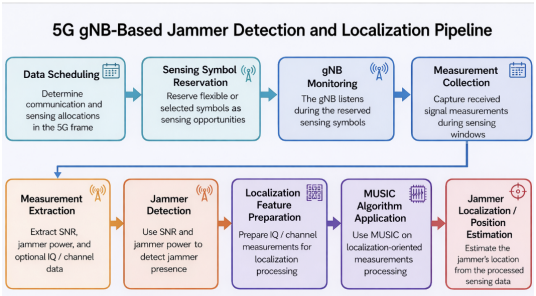


Fig. 2. System architecture and processing flow for the proposed sensing framework.

B. Data-Scheduling-Based Sensing Design

Instead of relying on Sounding Reference Signal (SRS) configuration sweeps, the updated framework introduces sensing opportunities directly through data scheduling within the 5G NR frame. The gNB scheduler was modified to blank symbols 12 and 13 in every tenth slot, creating a periodic quiet window during which the transmitter remains silent and the receiver captures ambient RF energy, interference, and jammer-induced signal components. This enables sensing without requiring additional hardware modifications.

The scheduling design was motivated by 3GPP TS 38.211 Table 4.3.2-1, in which symbols 12 and 13 may appear as flexible symbols in standard TDD slot formats. When configured as downlink quiet symbols, these intervals provide an opportunity for the gNB to observe the channel without expecting intended UE transmission. To support this feature, a new configuration structure was added to expose sensing-related parameters including the enable flag, quiet symbol start index, quiet symbol count, and sensing period in slots.

At 30 kHz subcarrier spacing, one slot has duration 0.5 ms. Since the sensing design reserves 2 symbols once every 10 slots, the sensing overhead is

$$\text{Overhead} = \frac{2}{10 \times 14} = \frac{2}{140} \approx 1.4\%.$$

This introduces only a small resource cost while creating periodic sensing opportunities within the normal communication process.

During each quiet sensing window, IQ samples are captured from the receive path and written to disk for post-processing. At the same time, a jammer proxy inserted into the ZMQ signal chain injects controlled continuous-wave interference and logs measurements such as UE power, jammer power, and signal-to-interference ratio. These synchronized measurements provide the input to the MATLAB-based detection stage described in the following subsection.

C. Detection and Localization Processing Framework

A post-processing framework was developed to analyze the sensing measurements captured during the scheduled quiet windows. IQ samples recorded from the modified srsRAN receive path and signal logs generated by the jammer proxy were converted into MATLAB-compatible data structures for downstream jammer detection. The main variables used in the current stage were time, measured SNR, jammer power, and a ground-truth jammer-state label indicating jammer OFF and jammer ON conditions.

To quantify separability between jammer OFF and jammer ON conditions, summary statistics were computed for the measured SNR and jammer power values. The sample mean and sample standard deviation were defined as

$$\mu = \frac{1}{N} \sum_{i=1}^{N} x_i \tag{1}$$

and

$$\sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2}. \tag{2}$$

Using the class means, the SNR and jammer-power thresholds were defined as

$$T_{\mathrm{SNR}} = \frac{\mu_{\mathrm{SNR,OFF}} + \mu_{\mathrm{SNR,ON}}}{2} \tag{3}$$

and

$$T_{\mathrm{JP}} = \frac{\mu_{\mathrm{JP,OFF}} + \mu_{\mathrm{JP,ON}}}{2}. \tag{4}$$

The SNR-based detector was defined as

$$\hat{y}_{\mathrm{SNR}}(i) = \begin{cases} 1, & \mathrm{snr_db}(i) < T_{\mathrm{SNR}} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

The jammer-power-based detector was defined as

$$\hat{y}_{\mathrm{JP}}(i) = \begin{cases} 1, & \mathrm{jammer_power}(i) > T_{\mathrm{JP}} \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

The combined detector was defined as

$$\hat{y}_{\mathrm{C}}(i) = \begin{cases} 1, & \mathrm{snr_db}(i) < T_{\mathrm{SNR}} \\ & \quad \vee\ \mathrm{jammer_power}(i) > T_{\mathrm{JP}} \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

The overall detection accuracy was computed as

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{8}$$

These statistics and decision rules were used to evaluate lightweight threshold-based jammer detection using the measured SNR and jammer power features.

To support the future localization stage, a separate experimental framework was also used to evaluate MUSIC-based angle-of-arrival estimation. This baseline employs a 3-element uniform linear array and spatial covariance matrix decomposition to separate signal and noise subspaces, followed by pseudospectrum evaluation for angle estimation. The MUSIC stage is not yet fully integrated into the srsRAN sensing path, but it provides a validated localization reference model for the next phase of the framework, consistent with prior high-resolution localization methods such as MUSIC [1] and SpotFi [2].

D. srsRAN Test Environment

To support experimentation within a realistic 5G communication framework, the full system was deployed on a Linux server using the srsRAN 5G software stack, Open5GS core network, and ZeroMQ (ZMQ)-based simulated radio links. The environment included a modified srsRAN gNB, an srsUE instance, and a jammer proxy inserted into the ZMQ signal path. This setup enabled controlled communication and jammer injection without requiring external RF hardware.

The system was configured for NR Band 3 with a center frequency of 1842.5 MHz, 20 MHz bandwidth, 30 kHz subcarrier spacing, single-antenna TDD operation, and a single-user scenario. The gNB and UE communicated normally during standard operation, while the jammer proxy introduced a configurable continuous-wave interference signal after a specified slot delay. Runtime execution was monitored through command-line logs and recorded signal measurements to verify correct gNB initialization, UE attachment, and jammer activation.

This test environment provided a controllable platform for validating the proposed data-scheduling-based sensing approach inside an open 5G implementation. It also supported synchronized capture of IQ data, jammer power, UE power, and signal-to-interference ratio for downstream MATLAB-based jammer detection and future MUSIC-based localization.

III. Results

A. Jammer Injection and Measurement Overview

Experiments were conducted to evaluate jammer detection using the proposed scheduler-based sensing pipeline. The jammer proxy injected a continuous-wave jammer with a 100 kHz frequency offset, amplitude of 0.8, and delayed activation at slot 2000, allowing the system to first establish a clean communication baseline. A total of 9454 sensing measurements were recorded, including 501 jammer OFF samples and 8953 jammer ON samples, as summarized in Table I.

Before jammer activation, corresponding to slots 0 through 1999, the average measured SNR was approximately 11.07 dB. After activation, the measured SNR dropped sharply to approximately -91.10 dB, indicating severe degradation in communication quality. At the same time, the measured jammer power increased from 0 to approximately 0.64. These feature changes establish a clear distinction between jammer OFF and jammer ON conditions and motivate the threshold-based detection results presented below.

TABLE I
DATASET STATISTICS UNDER JAMMER OFF AND JAMMER ON CONDITIONS.

Metric	Value
Total samples	9454
Jammer OFF samples	501
Jammer ON samples	8953
SNR mean (OFF)	11.07 dB
SNR std. (OFF)	31.23 dB
SNR mean (ON)	-91.10 dB
SNR std. (ON)	26.34 dB
Jammer power mean (OFF)	0.00
Jammer power std. (OFF)	0.00
Jammer power mean (ON)	0.64
Jammer power std. (ON)	0.00

B. Feature Behavior Over Time

Figure 3 shows jammer power over time, separated by jammer state. The jammer OFF samples remain clustered near zero, whereas jammer ON samples remain near 0.64 with essentially no visible overlap. This indicates that jammer power is a highly discriminative sensing feature in the current dataset.

Figure 4 shows the measured SNR over time. During normal operation, the SNR remains high and positive. Once jammer activity begins, the SNR collapses to strongly negative values, with intermittent spikes caused by repeated UE reconnection attempts. Compared with jammer power, SNR still reflects jammer activity clearly, but exhibits greater variation.

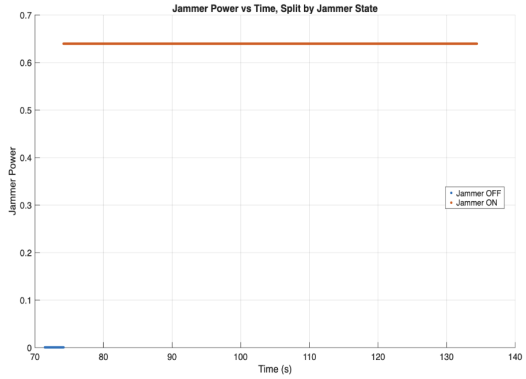


Fig. 3. Jammer power versus time for jammer OFF and jammer ON conditions.

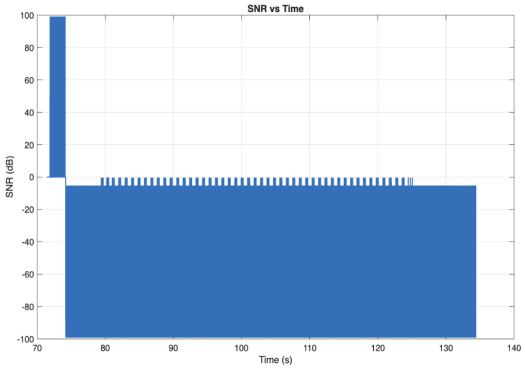


Fig. 4. SNR versus time under jammer sensing conditions.

C. Threshold-Based Jammer Detection

To quantify detection performance, three threshold-based detectors were evaluated using the measured SNR and jammer power features. Their numerical performance is summarized in Table II.

TABLE II
PERFORMANCE OF THRESHOLD-BASED JAMMER DETECTORS.

Detector	Threshold	Accuracy	TP	TN	FP	FN
SNR-based	-40.02 dB	92.18%	8214	501	0	739
Jammer-power-based	0.320	**100%**	8953	501	0	0
Combined	N/A	**100%**	8953	501	0	0

Figure 5 shows the SNR-based detector. The jammer OFF and jammer ON SNR distributions are substantially separated, with the decision threshold indicated by the vertical red line. However, some spread remains near the threshold, so the classes are not perfectly separable. As reported in Table II, the SNR-based detector achieved 92.18% accuracy, with 8214 true positives, 501 true negatives, 0 false positives, and 739 false negatives. This indicates that SNR is informative for jammer detection, but less robust than jammer power in the current dataset.

Figure 6 shows the jammer-power-based detector. In contrast to SNR, the jammer-power distributions are completely separated by the threshold, with jammer OFF values clustered at 0 and jammer ON values clustered near 0.64. As shown in Table II, this detector achieved 100% accuracy with no false positives or false negatives. This provides the clearest evidence that jammer power is the strongest first-stage sensing feature in the current dataset.

A combined detector was also evaluated by classifying a sample as jammer ON if either the SNR fell below the SNR threshold or the jammer power exceeded the jammer-power threshold. As shown in Table II, the combined detector also achieved 100% accuracy. However, it did not improve upon the jammer-power-based detector alone, indicating that jammer power already contains sufficient information to fully separate the two classes in the present experiment.

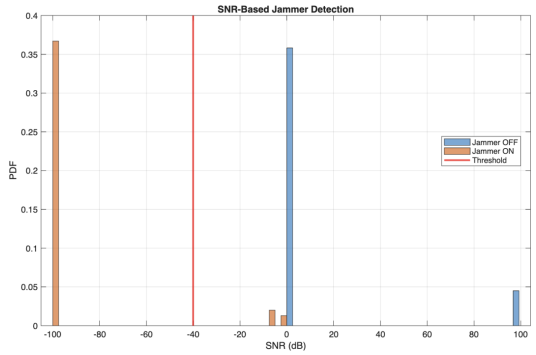


Fig. 5. SNR-based jammer detection histogram with threshold separating jammer OFF and jammer ON conditions.

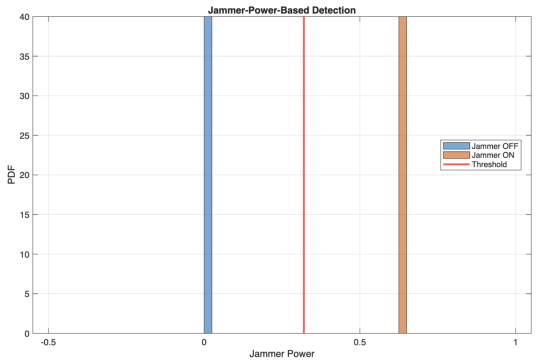


Fig. 6. Jammer-power-based detection histogram with threshold separating jammer OFF and jammer ON conditions.

D. MUSIC Localization Baseline

To support the future localization stage, a separate MATLAB simulation was used to validate a MUSIC-based angle-

of-arrival baseline. Figure 7 presents four diagnostics. The CSI phase across subcarriers remains approximately stable for all three antennas, indicating a well-conditioned input with preserved spatial phase structure. The I-Q constellation exhibits tight clustering, confirming acceptable baseband signal integrity. The covariance-matrix eigenvalue plot shows one dominant eigenvalue, consistent with a rank-1 single-source assumption. Finally, the MUSIC pseudospectrum shows a clear dominant peak aligned with the 25° ground-truth angle. Together, these results confirm that the MUSIC estimator successfully isolates the signal subspace and provides a valid localization baseline for future integration into the proposed jammer-sensing framework. This remains consistent with the role of MUSIC in the original manuscript as a high-resolution localization method [1].

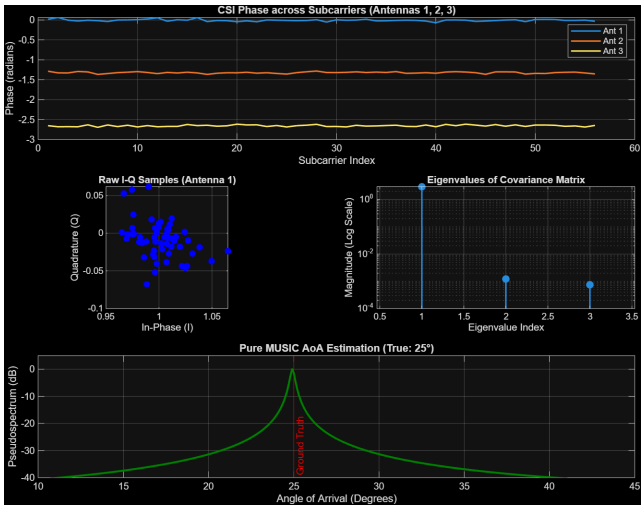


Fig. 7. Example of MUSIC-based angle-of-arrival estimation for future jammer localization.

IV. Discussion

The results show that jammer activity produces strong and consistent changes in the sensing features captured during the scheduled quiet windows. As summarized in Table I, the mean SNR decreases from 11.07 dB in the jammer OFF state to -91.10 dB in the jammer ON state, while the mean jammer power increases from 0.00 to 0.64. These shifts are also visible in Fig. 3 and Fig. 4, confirming that the proposed scheduler-based sensing approach can capture interference-related information directly from the communication process. This demonstrates that jammer sensing can be introduced with low overhead and without dedicated sensing hardware.

Among the evaluated features, jammer power was the strongest indicator of jammer presence in the current dataset. As shown in Fig. 6, the jammer OFF and jammer ON jammer-power distributions are cleanly separated by the selected threshold. This is reflected in Table II, where the jammer-power-based detector achieved 100% accuracy with no false positives and no false negatives. By contrast, the SNR-based detector in Fig. 5 achieved 92.18% accuracy. Although this still indicates strong detection performance, the nonzero false-negative count shows that SNR alone is less robust than jammer power under the present conditions. A likely reason is that SNR reflects overall communication quality and therefore exhibits greater variation than jammer power.

The combined detector achieved the same 100% accuracy as the jammer-power-based detector alone, as shown in Table II. This indicates that jammer power already contains sufficient information to fully separate the two classes in the current experiment, so adding SNR does not improve the result further. However, this does not imply that SNR is unimportant. In more realistic environments with fading, mobility, background interference, or more diverse jammer behaviors, SNR may still provide complementary information when jammer power is less cleanly separated.

These results should also be interpreted in light of the current experimental limitations. First, the dataset is imbalanced, with many more jammer ON samples than jammer OFF samples, as shown in Table I. Second, the ZMQ-based simulated radio environment introduces padding values into the receive buffer during certain quiet-symbol intervals, meaning that some captured IQ samples reflect simulation transport behavior rather than true ambient RF. This limitation does not invalidate the current detector evaluation, since the jammer proxy logs still provide synchronized ground-truth measurements for jammer activation and feature extraction, but it does mean that the present results should be interpreted as a controlled software testbed result rather than a full hardware-realistic sensing demonstration. A future implementation on USRP hardware would remove this artifact and allow the quiet sensing windows to capture actual ambient RF and jammer energy directly.

The MUSIC simulation result in Fig. 7 provides a baseline for the future localization stage. The stable CSI phase, clustered I-Q constellation, dominant covariance eigenvalue, and pseudospectrum peak aligned with the 25° ground truth indicate that the MUSIC estimator is correctly isolating the signal subspace. This is consistent with the role of MUSIC as a high-resolution angle-of-arrival method and supports its use as the next step after reliable jammer detection has been established [1]. The present work therefore validates the first stage of the overall framework: scheduled sensing measurements can be used for jammer detection, and future work can build on this foundation by integrating richer signal features and MUSIC-based localization into the same 5G communication pipeline.

Overall, the results demonstrate that communication-driven sensing opportunities created through scheduling are sufficient for reliable first-stage jammer detection in an open-source 5G framework. The current study shows that jammer power is the most discriminative feature in the evaluated dataset, while SNR remains a useful but less stable secondary feature. These findings validate the low-overhead sensing architecture and provide a practical foundation for extending the system toward near-real-time jammer localization.

The source code and related files to this paper can be found at this GitHub repository.

ACKNOWLEDGMENTS

This work was conducted as part of the Masters in Electrical and Computer Engineering (MECE) program at Rice University. The authors acknowledge the academic support and institutional resources provided by Rice University, as well as the guidance and feedback from the project advisor throughout the project. The authors also recognize the collaborative contributions of all team members in completing this research. The statements made in this work are solely the responsibility of the authors.

V. CRediT Author Statement

Sandra Mansour: Software, Formal analysis, Visualization, Writing–Original draft preparation. **Samanvitha Balijepalli:** Investigation, Algorithm analysis, Writing–Reviewing and Editing. **Aniket Jadhav:** Investigation, Methodology, Software. **Dr. Rahman Doost:** Conceptualization, Supervision, Project administration, Resources, Writing–Reviewing and Editing.

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